MDM4U GAMES PROJECT

Purpose:

The purpose of this project is to allow students the opportunity to demonstrate their understanding of overall course expectations.

Overall Course Expectations Covered

- Solve problems involving the probability of an event or a combination of events for discrete sample spaces
- Solve problems involving the application of permutations and/or combinations to determine the probability of an event
- Demonstrate an understanding of discrete probability distributions, represent them numerically, graphically, and algebraically, determine expected values, and solve related problems from a variety of applications

The Assignment

Students are to create an original, interesting, and easy-to-play game involving dice, spinners, cards, or any other reasonable item. The outcomes in your game **should ONLY depend on the element of chance.**

Part 1: Create the Game

- Develop a game and the rules
- Conduct experimental trials and record data (min. 50 plays of the game)
- Discuss and develop calculations for theoretical probability values
- Calculate and graph a probability distribution
- Calculate the expected value(use this to determine the cost of your game $1 or $2 data dollars)

Part II: Write a report of your Game

- Summarize how your game is played
- Show in detail how the probabilities of all outcomes are calculated for both theoretical and experimental data(probability distribution chart showing the calculations not just answers)
- Create graphs
- Interpret the graphs for both the theoretical and experimental probabilities

Things to look for…

- Use of experimental probability
    - Are the devices used (dice, coins, spinners, etc.) to simulate the event "good" devices (ie. Do they produce random outcomes? Verify!)
    - Playing the game over and over again (at least 50 times) and keeping track of important events
    - Graphs/diagrams/charts depicting the experimental probability (probability distribution) of the event(s) along with the expected outcomes
- Use of theoretical probability
    - Use of permutations and/or combinations where applicable
    - Use of Venn diagrams/tree diagrams/Pascal's triangle where appropriate
    - Use of probability distributions to aid your calculations
    - Use of graphs/diagrams/charts depicting the theoretical probability of the event(s)
- Analysis of the results of experimental and theoretical probabilities
    - Are they similar? Why or why not? A detailed explanation is required here.
- Determining the expected value/winnings for your game, experimentally and theoretically. o Is the game fair for all players? Justify your answer.
- Game strategies o Is there a winning strategy for playing your game?

Name: _____ Date: _____

Game Project Content Rubric

	Level 1 (50–59%)	Level 2 (60–69%)	Level 3 (70–79%)	Level 4 (80–100%)	Grade
Knowledge and Understanding (45)					
Introduction • description of the game & rules, relevant background information (5)	• Limited information about the game	• Sufficient information about the game	• Detailed description of most elements of the game	• Clear & detailed description of all elements of the game	
Data Collection • data table represents actual data from the game day or simulation (20)	• Insufficient data collected	• Sufficient data collected	• Sufficient data collected; data are appropriate to the task	• Extensive data collected; data are appropriate to the task & clearly represented	
Mathematical Understanding • probability calculations and data analysis (20)	• Reflects limited mathematical understanding	• Reflects some mathematical understanding	• Generally reflects mathematical understanding	• Reflects thorough mathematical understanding	
Application (45)					
Data Analysis • theoretical distribution • experimental distribution • expected value (45)	• Limited use of appropriate mathematical procedures and strategies	• Mathematical strategies and procedures generally appropriate and sometimes correct	• Appropriate strategies used most of the time • Mathematical procedures generally correct/appropriate	• Mathematical strategies & procedures always correct/appropriate and are highly detailed	
Thinking (30)					
Summary and Conclusions • compare/contrast theoretical and actual distributions • discuss discrepancies in the results • provide insights & reflections about the game & clear conclusions (20)	• Performs limited comparisons • Uses mathematical reasoning to justify few conclusions	• Performs some comparisons • Uses mathematical reasoning to justify some conclusions	• Performs detailed comparisons • Uses mathematical reasoning to justify all conclusions	• Performs and mathematically justifies conclusions based on insightful comparisons and reflections	
Self-Assessment • Examines characteristics of game for suitability • Gives consideration to the requirements and expectations of the project (10)	• Self-assessment is mostly incomplete or not complete	• Demonstrates some use of self-assessment to check progress and completeness of project	• Demonstrates significant use of self-assessment to check progress and completeness of project	• Carefully and thoroughly examines game design and analysis to make modifications and ensure project is thoroughly completed	

	Level 1 (50–59%)	Level 2 (60–69%)	Level 3 (70–79%)	Level 4 (80–100%)	Grade
Communication (30)					
Clarity • explanations clear, accurate, & thorough (12)	• Explanations are incomplete, inaccurate, and/or lack clarity	• Explanations are somewhat clear and accurate	• Explanations are mostly clear and accurate	• Explanations are clear, accurate, and thorough	
Math Terminology and Writing • mathematical terms used correctly • proper writing conventions followed (5)	• Limited use of correct terminology and notation • Proper writing conventions followed to a limited extent	• Some use of correct terminology and notation • Proper writing conventions followed sometimes	• Considerable use of correct terminology and notation • Proper writing conventions mostly followed	• Thorough and meticulous use of correct terminology and notation • Proper writing conventions extensively followed	
Organization • key ideas presented logically • appropriate headings (title page, table of contents, etc.) (5)	• Report not clear or logical; ideas rarely connect	• Report somewhat clear or logical; ideas connect sometimes	• Report most often clear, logical; ideas mostly connect	• Report clear, logical; ideas thoroughly connect	